

Q/R/M/C: A Stage-Decomposition Evaluation Protocol for Memory-Based Agents

Anonymous submission

Abstract

Success-only metrics collapse qualitatively different memory failures into a single scalar, hiding *which stage* of the retrieval pipeline is responsible. We contribute an evaluation protocol: the Q/R/M/C decomposition (Query formation, Retrieval satisfaction, target Materialization, Completion), operationalised so that four conditional rates are computable from standard trajectory logs. The arithmetic is the chain rule, deliberately so — like precision/recall, the value is identifiability and what the decomposition reveals. Single-agent (MiniGrid, BabyAI): oracle-stage interventions localise a retrieval-limited task ($0.39 \rightarrow 0.60$ under oracle retrieval) vs. downstream-limited ones, and log inspection refines the diagnosis — candidate *ranking* is near-optimal (96.9%) while candidate *generation* fails at 91% of decision points: a memory-accumulation limit. Multi-agent (35,640 runs, four sharing architectures, cadence K): the predicted bottleneck shift between Materialization and Completion is confirmed as cluster-robust slopes (Spearman $-0.53/+0.43$), time-to-success has an interior optimum at $K=8$, and a provenance annotation supplies the mechanism — the completion collapse of fast sharing concentrates in locks backed by foreign evidence. Externally, the identical tool-trace contract instruments three third-party agent stacks (OpenAI SDK, LangGraph, AutoGen): engineered structural failures localise identically on every stack ($R^*=0$, $C^*=0$ in 20/20), while on clean tasks the frameworks spread 0.50–0.75 end-to-end and the stages attribute the spread to weak first-commitment discipline with different recovery loops. The protocol makes memory failures comparable across symbolic, learned, LLM, single- and multi-agent systems.

Introduction

Embodied, goal-conditioned, and LLM agents increasingly retrieve places, objects, and facts *by meaning* from an explicit memory, yet are evaluated almost exclusively end-to-end: if success improves, memory is judged useful. Two systems with the same success rate can fail for entirely different reasons — one never accumulates queryable evidence, another retrieves perfectly and loses the target to a peer. This paper contributes the measurement instrument that separates such cases.

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The instrument is a four-stage decomposition of memory-based task completion: **Query** formation (the planner emits a semantic intent), **Retrieval** satisfaction (memory returns a candidate matching the task), target **Materialization** (the planner locks one candidate as an actionable target), and **Completion** (the controller reaches it). Logged as per-episode events Q^* , R^* , M^* , C^* , they factorise semantic-path completion into conditional rates estimable from ordinary trajectory logs (Definition 1; a multi-agent extension with memory and occupancy coupling channels is Definition 2). The consistency argument is the chain rule plus frequency estimation, and we claim no mathematical novelty for it: exactly as decomposing accuracy into precision and recall added no mathematics but changed what the field could see, the contribution is a *logging discipline* and what it reveals.

Contributions.

1. **The protocol.** Operational, log-level definitions of the four stage events for single agents (Definition 1) and communicating collectives (Definition 2), with identifiability under a stated conditional stage-Markov assumption, plus a stress test of that assumption.
2. **Diagnoses that success cannot make.** Single-agent: oracle-stage interventions and log forensics attribute failures to memory *accumulation* rather than ranking. Multi-agent: a registered slope-level claim (Empirical Claim 1) — fast sharing raises $P(M^*|R^*)$ and lowers $P(C^*|M^*)$ — is confirmed on a 35,640-run sweep, with an interior time-to-success optimum at $K=8$ and a provenance-level mechanism.
3. **External validity.** The same contract, defined once over tool traces, instruments OpenAI-SDK, LangGraph, and AutoGen agents without per-framework adaptation; engineered structural failures localise identically on all stacks, and behavioural differences between stacks are attributed stage-by-stage.

Table 1: The paper’s argument in miniature: one logging contract, five system classes, a located diagnosis in each. Every section instantiates one row.

System class	What the same four events revealed
Symbolic single agent	ranking 96.9% vs. empty candidates 91%: an <i>accumulation</i> limit
Symbolic collective	$M \leftrightarrow C$ bottleneck shift; t_{succ} optimum $K=8$; provenance mechanism
Learned policies	competitive success, collapsed stage observability
LLM agents	$1.00 \rightarrow 0.00$ collapse = pure C-budget; sharing lowers success $2/3 \rightarrow 1/3$
Third-party $\times 2$ models	structural failures localize identically; M_1 separates models; frameworks reshape it

Scope and non-claims. (i) No mathematical novelty in the factorisation; the contribution is operationalisation. (ii) Empirical Claim 1 is a verified slope-level observation, not a theorem. (iii) No claim about the conditional product Φ (monotone in our data); the interior optimum lives on time-to-success. (iv) Primary substrates are grid worlds; a continuous-state check (VMAS) is directional; LLM studies use local models at modest n . (v) Multi-agent place identity assumes a shared coordinate frame; a companion manuscript removes it. (vi) The protocol unifies *observation points*, not representations — but R^* and M^* presuppose an entity-correspondence oracle between memory content and world referents (here supplied by instrumented environments); without it the protocol degrades gracefully to Q^* and C^* . Grounding that correspondence is open.

The Q/R/M/C Protocol

An episode issues a semantic query q (required/preferred/penalty tags); memory returns candidates; the planner locks one; the controller executes. The four events anchor at logged triggers: Q^* fires on intent emission; R^* when the candidate set contains a task-satisfying candidate; M^* on lock acquisition of a true target; C^* on arrival. Per-episode flags are ORs over ticks; conditional rates $P(R^*|Q^*)$, $P(M^*|R^*)$, $P(C^*|M^*)$ are frequency estimates over episodes.

Definition 1 (single-agent estimability, informal). Under a conditional stage-Markov property — each stage outcome depends on the trace only through its predecessor stage — semantic-path completion factorises as $P(C^*) = P(Q^*)P(R^*|Q^*)P(M^*|R^*)P(C^*|M^*)$, and each factor is consistently estimable from logs. A simulation stress test quantifies estimator bias when the assumption is violated by a hidden confounder: $P(M^*|R^*)$ stays unbiased while $P(C^*|M^*)$ inflates, so diagnoses that hinge on the M-C contrast degrade gracefully (supplement).

Definition 2 (multi-agent extension, informal). With N agents sharing memory by peer broadcast every K ticks, the conditioning sets enlarge with two coupling channels: the shared memory $\mathcal{M}_{-i}^{(K)}$ and occupancy $\mathcal{O}_{-i}$ (scarce targets are consumed). The same four rates are logged per agent.

Empirical Claim 1 (bottleneck shift, slope-level). As K decreases (faster sharing), $P(M^*|R^*)$ rises — fresher peer evidence enables reliable commitment — and $P(C^*|M^*)$

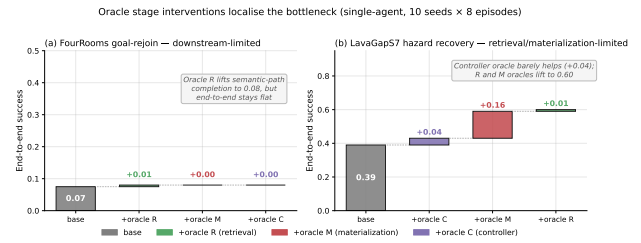


Figure 1: The diagnose→intervene→verify loop on the single-agent benchmark: oracle-stage interventions lift exactly the stage the decomposition indicates. Hazard recovery is retrieval-limited (oracle-R: 0.39 $\rightarrow$ 0.60 end-to-end); goal-region rejoin is downstream-limited (oracle-R rescues semantic-path completion, end-to-end unchanged).

falls — peers race for the same targets. Both are claims about conditional rates, not about their product.

Interpretive readings of the stages (an order-theoretic account of empty retrieval via the query–extent Galois connection; an options/semi-MDP account of why optima appear on time-bearing metrics) are deliberately confined to the supplement: they are lenses, not contributions.

Single Agent: Localizing the Failure

On a 10-seed MiniGrid benchmark (water/rest semantic search, goal-region rejoin in FourRooms, hazard recovery in LavaGapS7; BabyAI portability in the supplement), end-to-end success hides heterogeneity: water and rest are strong (0.95, 0.93), rejoin and hazard recovery are weak — for *different reasons*, which oracle-stage interventions separate. Replacing retrieval with an oracle lifts hazard-recovery success 0.39 $\rightarrow$ 0.60 (retrieval-limited); the same oracle rescues semantic-path completion on goal-region rejoin *without* changing end-to-end success (downstream-limited). The framework’s own logs then refine the attribution: when candidates exist, ranking precision is 96.9%; but the candidate set is *empty* at 91% of decision points. The binding constraint is not choosing among memories — it is *accumulating* queryable memory at all. This single number redirected our engineering from rankers to consolidation, and it is invisible to success-only evaluation.

A learned behaviour-cloning baseline makes the observability point: competitive success with collapsed stage structure (no explicit query/lock interface), so end-to-end optimisation trades away exactly the events the protocol audits.

Multi-Agent: Memory as a Shared Resource

Four sharing architectures over one symbolic substrate — independent private memories, a shared event bus, a central trust-weighted aggregator, and peer-to-peer broadcast at cadence $K \in \{1, \dots, 64\}$ — run a scarce water-search task ($N \in \{3, 5, 8\}$ agents, $T \leq N$ targets, three layouts, three hazard densities, 40 seeds; 35,640 runs). Figure 2 shows the stage profile of every architecture; retrieval saturates, and all differentiation lives on the M and C links, as Empirical Claim 1 predicts.

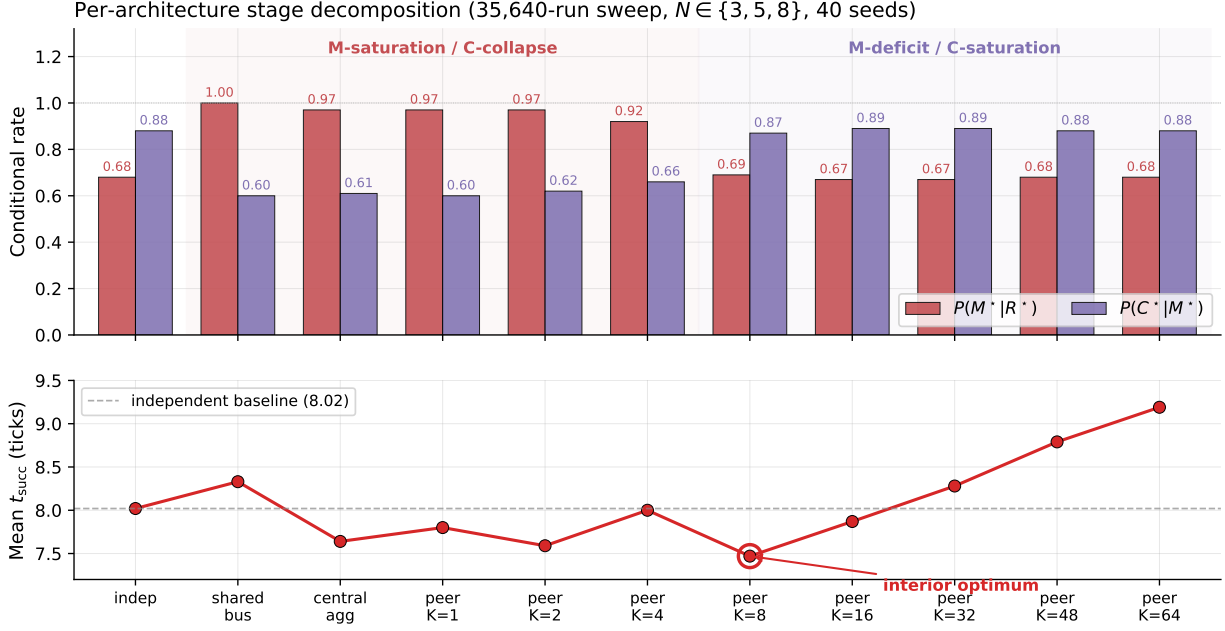


Figure 2: Per-architecture conditional Q/R/M/C rates (top) and mean time-to-success (bottom) on the 35,640-run sweep. Shared bus, central aggregator, and fast peer broadcast ($K \leq 4$) sit in the M-saturation/C-collapse regime; independent and slow peer ($K \geq 16$) in the reverse. Time-to-success has an interior minimum at $K=8$ (7.47 vs. 8.02 independent), robust in location, shallow in margin.

Bottleneck shift. Across peer runs, Spearman of $P(M^*|R^*)$ vs. K is -0.53 (cluster-robust 95% CI $[-0.59, -0.47]$ over 81 configuration cells) and of $P(C^*|M^*)$ vs. K is $+0.43$ ($[+0.37, +0.50]$). The sharper signature is within-cadence: Pearson between the two links inside a fixed K peaks at -0.61 at $K=4$ and decays to noise by $K=64$ — the bottleneck migrates between stages within the same configuration as K varies. Scaling to $N=12$ sharpens the signature (peak -0.70 , migrating to $K=8$). On mean time-to-success the trade-off produces an interior optimum at $K=8$; consolidation rate is a real, tunable design axis. Oracle interventions close the loop: at fast cadence, oracle-R/M/C progressively remove the excess time, attributing it to the $M \leftrightarrow C$ transition.

The mechanism is provenance. Annotating each M^* with the foreign share φ of its supporting evidence mass (computed from the merge’s own weights, frozen at lock time) and calling locks with $\varphi \geq 0.8$ warps: the warp share falls from 0.55 at $K=1$ to 0.11 at $K=8$ while own-evidence locks complete at a flat 0.26–0.34 and warp locks below 0.05 throughout. The C-collapse of fast sharing concentrates in the foreign-evidence stratum — cadence controls how much of the collective’s commitment runs on peers’ evidence, and that is the stratum that loses the occupancy race. The same curve reproduces on a continuous VMAS bridge ($0.72 \rightarrow 0.00$) and on an LLM text substrate ($0.42 \rightarrow 0.08$).

Learned and LLM collectives. A CommNet-PPO baseline matches symbolic peer success (0.667 vs. 0.65) with structurally degenerate stages (its real-valued channel is never informatively empty; R/M collapse) — competitive

success, zero stage observability. A collective of three LLM agents (llama3.1) sharing symbolic memory reproduces the phenomenology on language: pairing against a no-communication baseline on identical layouts, raw sharing lowers success $2/3 \rightarrow 1/3$ (occupancy racing), and a decentralized reservation-and-rollback protocol recovers part of it ($0.33 \rightarrow 0.42$) with rollback latency $\approx K$. On a step-budget sweep with two local models, an opaque success collapse ($1.00 \rightarrow 0.00$) resolves to pure C-stage failure ($Q^*=R^*=M^*=1, C^*=0$) identically for both models: a structural, not model-specific, diagnosis.

The Contract on Third-Party Agent Stacks

Does the protocol work on systems we did not build? We instrument three stacks — an OpenAI-SDK function-calling loop, LangGraph’s ReAct agent, and AutoGen — with the identical contract defined once over *tool traces*: an episodic household memory behind *recall/goto/take* tools; Q^* on querying, R^* on attribution (a returned record names the item at its true place), M^* on an episode-level lock, C^* on completion within a tool budget. Four variants engineer one failure each; all predictions were registered before the runs, and two revisions of the event semantics are recorded in the registration.

Three findings (Table 2, Fig. 4). (1) **Structural localizations are exact and framework-independent:** no consolidated fact $\Rightarrow R^*=0.00$; budget below requirement $\Rightarrow C^*=0.00$ at $Q^*=R^*=1$ — 20/20 episodes on every stack, no per-framework adaptation. (2) **The clean control separates the stacks, and the stages say how.** Success on identi-

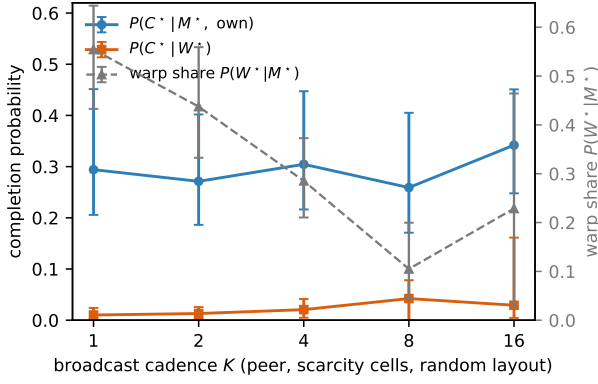


Figure 3: Provenance stratification of $P(C^*|M^*)$ vs. cadence K (peer, scarcity cells, episode-cluster bootstrap CIs). The share of locks backed by foreign evidence (warp share, grey) grows as sharing accelerates, while that stratum’s completion (orange) stays below 0.05 and the own-evidence stratum (blue) stays flat: the C-collapse of fast sharing is concentrated in the foreign-evidence stratum.

Table 2: Q/R/M/C on third-party stacks (20 episodes/cell). M_1 : first commitment targets the true place.

Stack	Variant	Q^*	R^*	M^*	C^*	M_1
OpenAI SDK	control	1.00	1.00	.85	.75	.25
	consol. gap	1.00	.00	.50	.40	.25
	stale ambig.	1.00	1.00	.90	.30	.25
	tight budget	1.00	1.00	.25	.00	.25
Lang-Graph	control	1.00	1.00	.70	.50	.50
	consol. gap	1.00	.00	.20	.15	.10
	stale ambig.	1.00	1.00	.65	.60	.50
	tight budget	1.00	1.00	.10	.00	.10
Auto-Gen	control	1.00	1.00	.85	.55	.45
	consol. gap	1.00	.00	.30	.20	.15
	stale ambig.	1.00	1.00	.50	.45	.35
	tight budget	1.00	1.00	.15	.00	.15

cal clean episodes spreads 0.50–0.75. A leaderboard reports “different quality”; the stages show a shared pathology — first-commitment discipline is weak everywhere ($M_1=0.25$ –0.50: the first `goto` frequently ignores the agent’s own just-retrieved evidence) — plus different loop economics: episode-level M recovers to 0.70–0.85 by re-query, and the stacks differ in how much survives to completion (C/M from 0.88 down to 0.65). **(3) The stale distractor pulls, unevenly.** An outdated record captures the first commitment in 0.20–0.40 of ambiguous episodes, and its completion cost is framework-specific (OpenAI SDK: 0.75 $\rightarrow$ 0.30; the others absorb the detour). Registered behavioural predictions that assumed a clean control failed *because the stacks themselves fail a clean memory task 25–50% of the time* — finding (2), reported as registered failures with mechanisms.

Model robustness (registered before the runs). Repeating all 240 episodes with a second model (qwen3:4b) confirms the registered prediction *exactly*: both structural local-

izations transfer unchanged ($R^*=0.00$ under the consolidation gap and $C^*=0.00$ under the tight budget, 20/20 on every stack). The behavioural profile, by contrast, is a MODEL property that the stages make legible: qwen3:4b completes every clean episode ($C^*=1.00$ on all three stacks) with first-commitment discipline $M_1=0.90$ –0.95 where llama3.1 managed 0.25–0.50, and is barely distracted by the stale record (0.05–0.30) — *except* under AutoGen, which drops qwen’s M_1 to 0.40: the decomposition separates what the model contributes from what the framework destroys. Full second-model table in the supplement.

Related Work

Cognitive-map and semantic-map lines motivate explicit place abstraction (Tolman 1948; Gupta et al. 2017; Savinov, Dosovitskiy, and Koltun 2018); goal-conditioned RL conditions on goals or embeddings (Andrychowicz et al. 2017); multi-agent RL studies learned channels (Sukhbaatar, Szlam, and Fergus 2016; Foerster et al. 2018) and gossip protocols (Boyd et al. 2006); LLM-agent systems add explicit memory layers evaluated end-to-end (Wang et al. 2024; Shinn et al. 2023; Packer et al. 2023; Park et al. 2023; Wu et al. 2023). The protocol is orthogonal to all four lines: it does not propose an architecture; it makes any of them stage-diagnosable from logs, and yields a slope-level cadence prediction testable on those logs.

Limitations

Primary substrates are grid worlds; VMAS is a directional continuous check; the LLM studies use small local models at modest n ; place identity across agents assumes a shared coordinate frame (removed constructively in a companion manuscript); Empirical Claim 1 remains slope-level — a derivation under a generative model would make it a theorem. Full tables, proofs of the estimability arguments, assumption stress tests, effect-size methodology, scale-up, and all registered-prediction files are in the supplement.

Conclusion

Q/R/M/C turns “the agent’s memory failed” from a scalar verdict into a located, comparable, and sometimes repairable diagnosis: a consolidation limit in a single agent, an occupancy-priced sharing rate in a collective, a commitment-discipline gap in off-the-shelf agent stacks. The protocol costs a logging contract, not a new architecture — and the same four numbers read across symbolic, learned, and LLM systems.

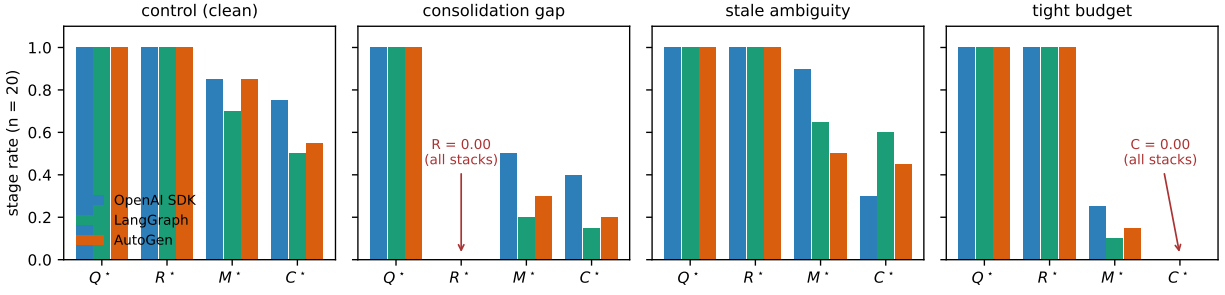


Figure 4: Stage profiles of three third-party stacks on four engineered variants (llama3.1, $n=20/\text{cell}$). Q and R saturate identically; the structural failures give exact zero columns on every stack; the inter-framework spread lives in M and C.

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